



The Perils Of Trivialising The IoT Security

This article is based on a talk by Sumanth Naropanth at India Electronics Week 2019 organised by EFY Group, where he talked about whose responsibility it is to secure data



Sumanth Naropanth is founder and chief executive officer, Deep Armor

Broadly, the Internet of Things (IoT) refers to a network of Internet-connected devices that are capable of gathering and sharing electronic information. It could be consumer goods, industrial equipment, traffic lights, construction equipment, healthcare devices and so on.

Data is the blood of IoT devices and, hence, it is important to manage and protect it. But how do you ensure integrity of data, its management or whether things are interconnected? These are some of the complexities and problems one may face from security as well as functional points of view.

There are a number of IoT protocols including ZigBee, Zwave, radio frequency (RF) and 4G, which means multiple options to choose from. So the question arises, is having multiple standards way

too many? Or, are too many standards an indication of having no standards that are good enough for security purposes?

The IoT ecosystem: an engineer's viewpoint

If you look at the IoT ecosystem from an engineer's perspective, there are four parts to it, namely, devices, mobile (which may or may not exist in some form factors), cloud and all networking aspects of it, including protocols.

Form factor devices (gateways, nodes, Edge, etc) are small devices of varying sizes. Depending on functionality, these might collect just sensor data and sensor values. Among these, gateways have a rich stack in the sense that these send all collected information back to the cloud. However, architectures may vary depending on use-